

- 1 (a) (i) Gravitational force provides the centripetal force

$$\frac{GMm}{r^2} = m\omega^2 r$$

$$\omega^2 = \frac{GM}{r^3}$$

$$\omega = \sqrt{GM} r^{-\frac{3}{2}}$$

$$k = -\frac{3}{2} = -1.5$$

- (ii)

$$\frac{\omega_2}{\omega_1} = \left(\frac{r_2}{r_1}\right)^{-\frac{3}{2}}$$

$$\omega_2 = \left(\frac{2\pi}{24 \times 3600}\right) \left(\frac{6700}{42000}\right)^{-\frac{3}{2}}$$

$$\omega_2 = 1.14 \times 10^{-3} \text{ rad s}^{-1}$$

- (b) (i)

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow mv^2 = \frac{GMm}{r}$$

$$E_k = \frac{mv^2}{2} = \frac{GMm}{2r}$$

$$E_T = E_K + E_P$$

$$= \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right)$$

$$= -\frac{GMm}{2r}$$

- (ii)

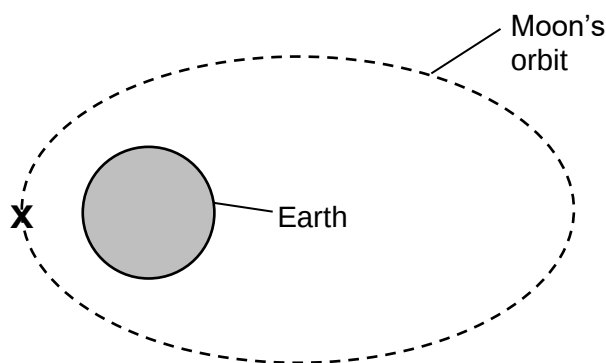
$$\Delta E_T = -\frac{GMm}{2r_2} - \left(-\frac{GMm}{2r_1}\right)$$

$$= \frac{GMm}{2} \left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$= \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(700)}{2} \left(\frac{1}{4.2 \times 10^7} - \frac{1}{(3.0 \times 10^5 + 4.2 \times 10^7)}\right)$$

$$= 2.37 \times 10^7 \text{ J}$$

- (c)



$$E_T = E_P + E_K$$

(E_T and E_P are negative values, E_K is positive.)

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E_T is constant, when Moon is nearest to the Earth, its E_P is smallest (more negative), hence E_K and speed is the largest.

OR By conservation of energy, when Moon is nearest to the Earth, its E_P is smallest (more negative), hence E_K and speed is the largest.